

show dimensions or special fabrication notes. A powerful scripting language, Lua, backs up the advantages of Osmond.

A lazy man's shortcut. Tools to move and reorient parts, connect and route traces, modify existing traces, and edit pad and pad stacks are available. Traces can be curved, and PCB labels can be made using characters from Mac font library. It is very easy to partially retrace an existing design or match the design with an existing mechanical drawing. To extend the capabilities, it supports circular, rectangular and oval pads, as well as variable trace widths and trace spacing.

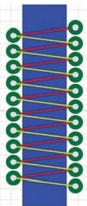
Flexibility of the tool. A flexible grid is provided to guide the work. Any PDF document can be imported and saved as a background image. This feature helps in retracing specific models much easily. A wide range of ground or inner planes using several methods is supported by Osmond. It also allows you to define arbitrary copper flood areas.

It is always the designer's way. Each time you start a new project, you can define the requirements with a variety of available options. Options to choose a number of layers, working areas width/height/margin, outer-layer trace width, inner-layer trace width, outer-layer spacing, inner-layer

Example: Lua Script for Current Transformer or Helical Antenna

The script has 13 parameters as listed below. Alterations in these parameters can be done to make a custom coil.

```
--Custom Coil Parameters
CTSecondaryCenterX = 50 * mm
CTSecondaryCenterY = 50 * mm
CTSecondaryTraceWidth = 6 * mil
CTSecondaryViaDiameter = 50 * mil
CTSecondaryHalfTurns = 20
CTSpacing = 6 * mil
CTSecondaryLayerA = 1
CTSecondaryLayerB = 3
CTPrimaryTraceWidth = 4 * mm
CTSecondaryDirection = 1 -- +1: clockwise, -1: counter clockwise
CTSecondaryRotation = 1 -- 1 for up to down, 0 for across
CTViaTypeName = "VIA" -- "VIA" is the OsmondPCB default via, you may create new via types and put its name here
CTViaNamePrefix = "CTVIA"
```



The coil shown above is designed for a four-layer board. The colours corresponding to top, second and third layers are red, blue and light-green, respectively. The script does not draw a primary (blue) trace. Parameters should be designed with care because the script assumes the width of secondary traces to be smaller in diameter. Spacing should be increased if traces are larger than via diameter.

spacing and solder mask ring are provided to help you design a PCB perfectly suiting your requirement.

Customisable, powerful predefined templates. Major and minor grids help create new designs. The main window is reserved for the PCB canvas, which contains a toolbar on top with basic tool buttons like select, zoom, rotate, attach traces, add pegs to a trace, wrap traces around pegs or pads, draw freehand traces, add text content, create pins, add elements from a library, select, swap and connect signals and more. Text based PCB files and library files in the software are easily readable by humans and programs.

Simple and efficient inbuilt

parts editor. A parts editor is integrated into Osmond, which helps to easily define new part types or modify existing parts. Part types can be either through-hole or surface mount. If a design from schematic capture application is brought into focus, Osmond imports part and net lists to maintain adherence to specified design rules throughout the process. Designs can also be completely informal, in an *ad-hoc* manner, without part or net lists at all. In other words, Osmond supports any design style.

Even experienced engineers sometimes make silly mistakes while designing, but an automated tool like Osmond and its supportive platform makes PCB designing a cakewalk. A variety of options provided by Osmond supports even the out-of-the-box ideas of designers, making the design flexible and efficient. **EFY**

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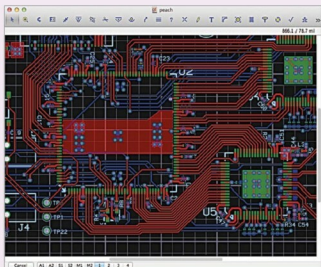


Fig. 2: PCB designing in Osmond